

Multimodal Mamba Fusion for Noise-Robust EEG Biometric Identification

Ly Ngoc Vu*, Huynh Cong Bang*, and Nguyen Minh Hanh[†]

Tóm tắt nội dung—Brain-derived signals offer a spoof-resistant biometric modality, but electroencephalogram (EEG) authentication remains brittle under environmental noise and stricter held-out-subject protocols. Building on our two-stage WaveNet+ResNet pipeline, we introduce a multimodal Mamba-augmented framework that (i) augments a WaveNet denoiser with a selective state-space (Mamba) block, (ii) adds a parallel spectrogram branch that performs frequency- and time-axis Mamba sweeps, and (iii) fuses the two views with a gated self-attention fusion module trained with an embedding-level mixer gate. We perform a systematic four-version comparison (V1: WaveNet only; V2: WaveNet+Mamba; V3: tuned Mamba quick run; V4: full multimodal with gated self-attention fusion) on the PhysioNet Auditory-Evoked-Potential dataset under a held-out-subject open-set protocol with session-disjoint closed-set retrieval (holdout subjects {2, 5, 7, 12}, three random seeds, three noise families). The proposed multimodal model attains the best Precision@1 on all three noise families (82.4% Gaussian, 87.7% power-line, 83.8% EMG); the corresponding Stage I reconstruction quality is SI-SNR 12.15/32.38/13.83 dB. We also expose a verification gap (AUROC $\approx$ 0.50) on held-out subjects that motivates calibration as future work.

Tín hiệu não bộ được xem như là một phương thức sinh trắc học giàu tiềm năng nhờ khả năng chống giả mạo cao; tuy nhiên, xác thực dựa trên điện não đồ (EEG) vẫn chưa đủ vững chắc trước nhiều môi trường và dưới các giao thức đánh giá nghiêm ngặt hơn trên các đối tượng chưa từng xuất hiện trong tập huấn luyện.. Tiếp nối quy trình gồm hai giai đoạn WaveNet+ResNet trước đây, bài báo đề xuất khung đa phương thức tích hợp khối Mamba: (i) bổ sung khối không gian trạng thái có chọn lọc vào bộ khử nhiễu WaveNet, (ii) thêm nhánh spectrogram song song với hai lượt quét Mamba theo trục tần số và thời gian, và (iii) hợp nhất hai biểu diễn bằng mô-đun gated self-attention fusion. Trên tập dữ liệu PhysioNet Auditory-Evoked-Potential với giao thức giữ kín người dùng (subject {2, 5, 7, 12}, ba seed, ba loại nhiễu), mô hình đề xuất V4 đạt Precision@1 cao nhất trên cả ba loại nhiễu (82.4% Gaussian, 87.7% nguồn điện, 83.8% EMG). AUROC $\approx$ 0.50 trên người dùng giữ kín cho thấy cần chuẩn hóa ngưỡng trong các nghiên cứu tiếp theo.

Index Terms—EEG biometrics, denoising autoencoder, state-space models, Mamba, multimodal fusion, gated self-attention fusion, metric learning.

I. INTRODUCTION

Manuscript submitted to the Student Scientific Research Communication (SSRC), 2026.

* L. N. Vu and H. C. Bang contributed equally to this work (co-first authors).

[†] N. M. Hanh is the corresponding author (e-mail: hanh_01036046@iuh.edu.vn).

All authors are with the Faculty of Information Technology, Industrial University of Ho Chi Minh City, Ho Chi Minh City, Vietnam (e-mail: dezzhug@gmail.com; bang_01036047@iuh.edu.vn; hanh_01036046@iuh.edu.vn).

THE push toward unobtrusive authentication for wearables, brain-computer interfaces, and cyber-physical systems has revived interest in EEG-based biometrics: the signal is internal, hard to spoof, and rich in subject-specific micro-patterns [12]–[14]. Concretely, we treat EEG biometrics as the following task: given a 4-channel, $T = 800$ -sample EEG window from an enrolled subject, possibly corrupted by deployment noise, output a unit-norm 128-d biometric embedding such that (a) the closest enrolled identity in cosine distance is the correct subject (closed-set retrieval), and (b) an unseen subject can in principle be flagged via a distance threshold (open-set verification). Two practical challenges persist: real recordings are corrupted by power-line interference, muscle artifacts, and broadband noise [11], and deployable systems must generalize to subjects that were never seen during training [6], [7].

Our prior two-stage pipeline [1] decouples a WaveNet denoiser from a ResNet metric-learning embedder but operates on a single view of the signal with a fixed convolutional receptive field. This paper grounds three contributions in concrete observations on that design. (i) The denoiser’s receptive field is fixed by depth and dilation, so we insert a *selective state-space* (Mamba) [3], [4] block at the midpoint of the WaveNet stack for content-aware mixing in linear time. (ii) The raw waveform hides rhythmic line-noise structure, so we add a *spectrogram branch* that re-encodes the denoised signal into a 2-D time-frequency map with frequency- and time-axis Mamba sweeps. (iii) A single embedding view risks branch-specific bias, so we fuse the two embeddings with a *gated self-attention fusion* module, jointly trained with auxiliary metric losses on each branch.

We evaluate four progressively richer architectures on the PhysioNet Auditory-Evoked-Potential EEG dataset [11]: V1 is the WaveNet-only denoiser baseline of [1]; V2 inserts a Mamba block at the WaveNet midpoint; V3 is a single-seed H100 quick run with a tuned Mamba preset, reported as a directional reference; V4 is the full multimodal proposal. Under a held-out-subject open-set protocol with session-disjoint closed-set retrieval, V4 delivers the highest mean closed-set Precision@1 on every noise family in our benchmark and reaches up to 32.4 dB SI-SNR on power-line noise.

II. RELATED WORK

EEG biometrics. Early EEG biometric systems relied on hand-crafted spectral features [12]. Deep models then took over, including attention-recurrent architectures [14], triplet-loss embeddings on raw signals [13], and Siamese networks for cross-session matching [12]. Compact convolutional networks

such as EEGNet [17] and deep ConvNets for EEG decoding [19] provide widely used backbones, but they are generally trained for closed-set classification and do not explicitly target open-set verification.

Denoising and signal enhancement. WaveNet-style dilated convolutions are a strong, well-studied tool for raw-waveform denoising [2], [10]. Our prior work [1] adopts this denoiser with an SI-SNR objective; the current paper investigates whether replacing part of the convolutional pathway with a selective state-space block produces a qualitatively cleaner signal.

State-space models. Structured state-space sequence models (S4) [4] and the more recent Mamba [3] approximate long-range temporal dependencies in linear time, with content-dependent gating. They are attractive for biomedical time-series where Transformer attention is costly [15] and where physiological non-stationarity makes hand-tuned filters fragile. EEG-specific Mamba variants have begun to appear for cross-subject decoding [18]; to our knowledge, ours is the first study that combines a Mamba-augmented denoiser with a Mamba spectrogram branch and gated self-attention fusion for EEG biometric identification with open-set verification analysis.

Metric learning and multimodal fusion. Angular margin and pair-weighted losses (ArcFace [6], multi-similarity [7], contrastive [9], triplet [8]) are standard tools for open-set verification. Audio spectrogram transformers [16] demonstrate that converting a 1-D acoustic signal into a 2-D time-frequency map and applying sequence models on top is effective; we adopt the same intuition with Mamba sweeps along both spectrogram axes and use a gated self-attention fusion to combine the two views, instead of a simpler concatenation or score-level fusion.

III. METHOD

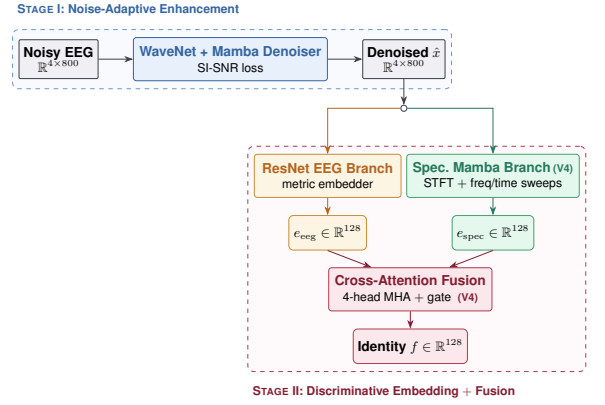
A. Overall Pipeline

The pipeline keeps the two-stage decomposition introduced in [1] but enriches each stage. Stage I is responsible for removing deployment noise from the raw waveform, and Stage II is responsible for mapping the cleaned signal to a 128-d biometric embedding. Stage I is a WaveNet+Mamba denoiser trained with a scale-invariant signal-to-noise ratio (SI-SNR) loss; Stage II is a multimodal embedder that consumes the denoised waveform together with a short-time Fourier transform (STFT)-based spectrogram and emits a 128-d L2-normalized biometric embedding (the L2-normalization layer is what produces the unit-norm vectors used downstream). The two stages are trained sequentially: Stage I for 30 epochs and then frozen as the upstream module for Stage II.

Fig. 1 sketches the overall flow, and Fig. 2 details the V4 multimodal architecture.

B. Stage I: Noise-Conditioned Waveform Denoiser

The denoiser ingests a 4-channel raw EEG epoch $x_{\text{noisy}} \in \mathbb{R}^{4 \times 800}$ and emits a clean estimate $\hat{x} = \mathcal{D}(x_{\text{noisy}})$. The WaveNet backbone contributes multiscale dilated convolutions for raw-waveform modeling, while the inserted Mamba block contributes selective, content-aware long-range mixing



Hinh 1. Two-stage decomposition: a noise-adaptive enhancement stage feeds a discriminative embedding stage. V4 augments the second stage with a spectrogram branch and gated self-attention fusion.

in $\mathcal{O}(L)$ operations. We retain the dilated-convolution backbone of [2], [10]: three blocks of four gated dilated layers with dilation 2^i for $i \in \{0, 1, 2, 3\}$. At the midpoint of the stack we insert a single *Mamba block* that performs content-dependent state-space mixing over the temporal axis with $d_{\text{model}} = 64$, $d_{\text{state}} = 16$, kernel 4, and expand factor 2. A pre-LayerNorm and a residual connection wrap the block. Architecturally, this swaps a single fixed-receptive-field convolutional layer for a selective state-space layer that re-weights information across the entire epoch in $\mathcal{O}(L)$ operations.

We synthesize paired training data, i.e. (noisy input, clean target) EEG pairs, by mixing clean EEG with three noise families: Gaussian, 50 Hz power-line, and broadband EMG-like (20–80 Hz) at controlled SNR levels. The training objective is the SI-SNR loss, a scale-invariant reconstruction-quality indicator standard in source separation [10], used here so that Stage I is robust to amplitude rescaling between the clean target and the reconstructed estimate. The denoiser is trained for 30 epochs:

$$\mathcal{L}_{\text{SI-SNR}}(\hat{x}, x_{\text{clean}}) = -10 \log_{10} \frac{\|\alpha x_{\text{clean}}\|^2}{\|\hat{x} - \alpha x_{\text{clean}}\|^2}, \quad (1)$$

$$\alpha = \frac{\langle \hat{x}, x_{\text{clean}} \rangle}{\|x_{\text{clean}}\|^2}.$$

C. Stage II: Dual-View Biometric Embedder

EEG branch. The denoised epoch is reshaped from $(4, 800)$ to $(4, 25, 32)$ and passed through a ResNet-18 or ResNet-34 backbone [5] with a 3×3 stride-1 stem and no maxpool to preserve the small spatial structure, followed by a projection head $\text{Linear}(\cdot, 256) \rightarrow \text{ReLU} \rightarrow \text{Dropout}(0.1) \rightarrow \text{Linear}(256, 128) \rightarrow \text{BatchNorm}$ and L2 normalization, producing the unit-norm waveform embedding $e_{\text{eeg}} \in \mathbb{R}^{128}$.

Spectrogram branch. We compute a short-time Fourier transform of the denoised signal with $n_{\text{fft}} = 128$ and hop length 64, yielding a 4-channel time-frequency tensor $S \in \mathbb{R}^{4 \times 65 \times 13}$. A two-layer convolutional stem ($4 \rightarrow 32 \rightarrow 64$ channels with 3×3 kernels and BatchNorm-ReLU) projects S into a 64-d feature map. We then apply two Mamba sweeps with a shared hidden width of 64: *frequency-axis* (sequence

is the frequency axis with the time axis flattened into the batch) and *time-axis* (sequence is the time axis with frequency flattened). After mean-pooling over the spatial dimensions, an MLP head produces the unit-norm spectrogram embedding $e_{\text{spec}} \in \mathbb{R}^{128}$ via a final L2-normalization layer that is what makes the embedding unit-norm.

Gated self-attention fusion. The two unit-norm vectors are stacked into a length-2 token sequence $T = [e_{\text{eeg}}; e_{\text{spec}}] \in \mathbb{R}^{2 \times 128}$ and passed through a four-head self-attention block (shared $Q/K/V$ derived from T ; referred to as a cross-attention module in [1]). The attended tokens $\tilde{T} = [\tilde{e}_{\text{eeg}}; \tilde{e}_{\text{spec}}]$ are mixed by a learned gate $g = \sigma(W_g[e_{\text{eeg}}; e_{\text{spec}}]) \in (0, 1)^{128}$ computed from the pre-attention embeddings, where σ is the elementwise sigmoid and $W_g \in \mathbb{R}^{128 \times 256}$ is a learned linear projection acting as a soft per-coordinate selector between the two views,

$$e_{\text{mix}} = g \odot \tilde{e}_{\text{eeg}} + (1 - g) \odot \tilde{e}_{\text{spec}}, \quad (2)$$

followed by a residual MLP and L2 normalization to give the fused biometric embedding $f \in \mathbb{R}^{128}$ used at inference.

Joint training. Stage II is trained for 30 epochs with one of two metric losses on f : ArcFace [6] or multi-similarity [7], drawn from [20]. To prevent the fusion module from collapsing onto one branch, we add weighted auxiliary metric losses on each branch:

$$\mathcal{L}_{\text{II}} = \mathcal{L}_{\text{metric}}(f) + \lambda_e \mathcal{L}_{\text{metric}}(e_{\text{eeg}}) + \lambda_s \mathcal{L}_{\text{metric}}(e_{\text{spec}}), \quad (3)$$

with auxiliary-loss weights fixed at $\lambda_e = 0.3$ and $\lambda_s = 0.2$ throughout this paper. A class-balanced minibatch sampler (MPerClassSampler) draws $m = 4$ samples per identity per batch to guarantee enough positive pairs.

IV. EXPERIMENTAL SETUP

Dataset. We use the publicly available PhysioNet *Auditory-Evoked-Potential EEG-Biometric* corpus [11], recording 20 subjects across 12 sessions of approximately two minutes each under both resting-state and auditory-stimulation conditions, with in-ear and bone-conduction headphones. Auditory evoked potentials are the cortical response to a controlled acoustic stimulus; their latency- and amplitude-domain morphology can carry subject-specific structure that originates in individual differences of cortical anatomy and auditory processing pathways, which makes them attractive as an internal, replay-resistant biometric cue. Recordings are segmented into fixed-length 4-channel epochs of $T = 800$ samples.

Evaluation protocol. We adopt a hybrid protocol that separates two evaluation tasks. (i) For *closed-set retrieval* (P@1, P@5), the non-holdout subjects are split per-subject into train, validation, and test sets, so test queries come from identities the embedder saw at training time but from unseen sessions or epochs (session-disjoint closed-set retrieval). (ii) For *held-out-subject open-set scoring*, the held-out subjects $S_{\text{holdout}} = \{2, 5, 7, 12\}$ are removed from training entirely and reused only at evaluation as “unknown” identities. We form a per-subject prototype as the mean of the training-set fused embeddings for that subject, score each held-out-subject epoch by its minimum Euclidean distance to any prototype,

and choose the operating threshold as the 95th percentile of the same min-distance scores measured on the validation split. The held-out-subject AUROC under this scoring stays close to chance (≈ 0.50) and is reported in the discussion rather than in Table II; the AUROC and EER columns in Table II are the closed-set pairwise verification scores (cosine similarities over all test-pool embedding pairs, session- and epoch-disjoint) and characterize within-subject session variability rather than held-out-subject novelty. We deliberately avoid the “strict subject-disjoint identification” label since identification queries are not from unseen subjects. We repeat each configuration with three random seeds (PyTorch, NumPy, and Python random seeds set to $\{0, 1, 2\}$) and report mean $\pm$ standard deviation.

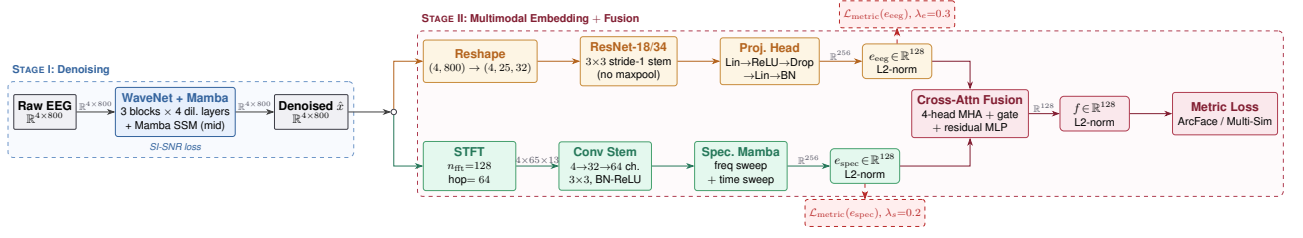
Noise families. We synthesize three noise types at deployment: white Gaussian, 50 Hz sinusoidal power-line, and broadband 20–80 Hz EMG-like noise.

Architectures. We compare four variants under identical data, splits, seeds, and training schedules:

- **V1:** WaveNet denoiser only (no Mamba) + EEG-only metric embedder. This is the prior pipeline of [1] re-evaluated under our held-out-subject open-set protocol.
- **V2:** V1 with the midpoint Mamba block in the denoiser.
- **V3:** V2 with a tuned Mamba preset, run as a directional, single-seed, different-hardware quick run (NVIDIA H100 with `torch.compile`); not used for the controlled three-seed claims.
- **V4:** full multimodal model proposed in this paper: WaveNet+Mamba denoiser, ResNet EEG branch, Mamba spectrogram branch, and gated self-attention fusion with auxiliary branch losses.

Implementation. All three-seed V1, V2, and V4 experiments run on a single NVIDIA RTX 5090 with batch size 256, 8 data-loader workers, mixed precision, and the denoised spectrogram source for V4. The V4 model has 22.8M parameters and an end-to-end inference latency of 186–197 ms per 4 s epoch on an RTX 5090 (ResNet-34+ArcFace head), well below the epoch duration and therefore real-time-compatible. V3 was a single-seed quick run on an NVIDIA H100 with `torch.compile`, reported as directional only.

Metrics. We distinguish a *score* (a cosine similarity over a single pair of embeddings) from a *metric* (an aggregation over the score population). For closed-set retrieval we report Precision@1 (P@1) and Precision@5 (P@5): P@5 is the fraction of the top-5 retrieved gallery epochs (excluding the query itself) that share the query identity, averaged across queries. For signal quality we report SI-SNR in decibels. For closed-set pairwise verification we report AUROC and equal-error rate (EER): AUROC is the area under the receiver-operating-characteristic curve constructed by sweeping a threshold over the cosine-similarity score, and EER is the threshold at which false-accept and false-reject rates coincide. Both are computed over all test-pool embedding pairs (Table II); held-out-subject open-set scoring is described separately in the protocol paragraph.



Hinh 2. V4 multimodal architecture. The WaveNet+Mamba denoiser feeds two parallel branches: a ResNet metric embedder on the denoised waveform and a spectrogram Mamba branch with frequency- and time-axis sweeps. A gated self-attention fusion module fuses the two L2-normalized embeddings into a single 128-d biometric embedding. Auxiliary metric losses on each branch (dashed) prevent fusion collapse during joint training.

Bảng I

BEST PRECISION@1 (P@1) PER VERSION AND NOISE TYPE. HIGHER IS BETTER. V3 IS A SINGLE-SEED QUICK RUN.

Noise	V1	V2	V3 [†]	V4
Gaussian	0.822	0.798	0.749	0.824
Power-line	0.860	0.858	0.869	0.877
EMG	0.824	0.811	0.758	0.838

[†] Single-seed H100 quick run; reported for reference only.

Bảng II

V4 MULTIMODAL RESULTS, MEAN±STD OVER THREE SEEDS. THE RESNET-34+ARCFACE HEAD IS THE STRONGEST BY P@1 ON ALL NOISE FAMILIES.

Noise	Head	P@1	P@5	SI-SNR	AUROC	EER
Gauss.	R34+MS	.764±.050	.743±.053	12.15±.26	.443±.021	.376±.072
	R18+MS	.782±.028	.755±.035	12.15±.26	.441±.036	.369±.046
	R34+Arc	.824±.036	.803±.040	12.15±.26	.479±.095	.356±.059
P-line	R34+MS	.819±.040	.794±.049	32.38±1.06	.533±.051	.408±.024
	R18+MS	.831±.047	.806±.052	32.38±1.06	.450±.015	.371±.021
	R34+Arc	.877±.006	.855±.005	32.38±1.06	.501±.044	.360±.039
EMG	R34+MS	.764±.080	.747±.075	13.83±.35	.439±.065	.389±.039
	R18+MS	.793±.067	.775±.069	13.83±.35	.422±.038	.356±.066
	R34+Arc	.838±.057	.827±.065	13.83±.35	.514±.082	.350±.058

MS: multi-similarity loss; Arc: ArcFace; R18/R34: ResNet-18/34.

Bảng III

ARCHITECTURAL ABLATION ON RESNET-34+ARCFACE BY P@1. Δ ROWS ARE V4 MINUS V1. STAGE I IS RETRAINED INDEPENDENTLY PER VERSION, SO SI-SNR IS NOT USED HERE AS AN ABLATION METRIC.

Noise	Variant	P@1
Gaussian	V1: WaveNet	0.822
	V2: + Mamba	0.798
	V4: + Spec + Fusion	0.824
	Δ V4-V1	+0.002
Power-line	V1: WaveNet	0.860
	V2: + Mamba	0.858
	V4: + Spec + Fusion	0.877
	Δ V4-V1	+0.017
EMG	V1: WaveNet	0.824
	V2: + Mamba	0.811
	V4: + Spec + Fusion	0.838
	Δ V4-V1	+0.014

C. Ablation: Mamba and Multimodal Fusion

Table III isolates the contribution of each architectural ingredient on the strongest head (ResNet-34 + ArcFace) using P@1 as the discriminating metric. Inserting a single Mamba block into the WaveNet denoiser (V1→V2) leaves P@1 essentially unchanged and even slightly hurts it, suggesting that swapping a single fixed-receptive-field convolutional layer for a state-space block does not, on its own, re-shape the embedding geometry. Adding the spectrogram branch and gated self-attention fusion (V2→V4) is what unlocks retrieval gains of +2.6, +1.9, and +2.7 percentage points respectively over V2, and +0.2, +1.7, and +1.4 over V1. We do not attempt to attribute SI-SNR deltas to the Stage II additions: each version retrains its own Stage I denoiser independently, so cross-version SI-SNR differences reflect independent training runs of nominally the same architecture rather than any contribution from the spectrogram branch or fusion module. Per-version Stage I SI-SNR (mean across seeds, ResNet-34+ArcFace configuration) is V1: 10.64/19.98/11.65 dB; V2: 10.65/19.61/11.67 dB; V4: 12.15/32.38/13.83 dB on Gaussian / power-line / EMG, and is intended only as descriptive Stage I quality.

D. Discussion

Where multimodal helps. The largest mean-P@1 jumps occur on power-line noise, consistent with the spectrogram localizing the 50 Hz line and the gate suppressing it before

V. RESULTS AND DISCUSSION

A. Cross-Version Best P@1

Table I summarizes the best Precision@1 of each version, selected per noise family across the three metric heads. The proposed V4 model is the strongest configuration on every noise type, with the clearest margins on power-line and EMG. The single-seed V3 quick run produces a competitive power-line score but is noisier overall and is shown for completeness.

B. Detailed V4 Results

Table II reports the full V4 metric grid for the three metric heads. ResNet-34 with ArcFace is consistently the strongest head in P@1, while ResNet-18 with multi-similarity is competitive on power-line. The denoiser reaches its highest SI-SNR in the power-line condition (the 50 Hz sinusoid is structurally easy to remove); Gaussian and EMG SI-SNR are lower but still in the regime expected for non-stationary residuals.

fusion. EMG is intermediate (broadband but high-frequency-tilted); Gaussian gains are smallest, consistent with white noise being structureless in both views.

Magnitude observation. V4 over V1 mean-P@1 deltas are within across-seed standard deviation, reported as a magnitude observation rather than a tested claim (see Limitations).

Verification gap. Across all V4 configurations, the held-out-subject AUROC of the centroid min-distance score remains close to chance (≈ 0.50), confirming that threshold-free verification of unseen identities is the open problem. The AUROC (0.42–0.53) and EER (0.35–0.41) values in Table II are computed on the closed-set test pool (session- and epoch-disjoint, all subjects already seen during training) and should be read as a within-subject session-variability indicator; verification calibration is left to future work.

Why Mamba alone (V2) is not enough. V2 vs V1 mean-P@1 deltas are within across-seed standard deviation; V4 over V1 gains should be read as a property of the full multimodal stack rather than an isolated Stage-I Mamba effect.

Limitations and future work. The dataset is small (20 subjects, 4 channels) with synthetic noise injection and only partial cross-day modelling; real-artifact recording-environment validation is planned. AUROC near 0.5 indicates an under-optimized verification objective for which calibration is planned. We did not run loss-weight (λ_e , λ_s) or fusion-component ablations; a grid sweep over λ_e , λ_s , and m is planned. Paired significance tests across V1–V4 are deferred to a planned per-seed cross-variant P@1 dump. Attention-map, frequency-bin, and denoising-waveform visualizations and external EEG-biometric baselines (EEGNet, ChronoNet, ICA+SVM) under the same protocol are planned for the next experimental round.

VI. CONCLUSION

We presented a multimodal Mamba-augmented framework for noise-robust EEG biometric identification extending [1]: V4 combines a WaveNet+Mamba denoiser, a spectrogram Mamba branch, and a gated self-attention fusion module trained with auxiliary branch losses. Under a held-out-subject open-set protocol with session-disjoint closed-set retrieval on the PhysioNet AEP dataset, V4 attains the highest mean Precision@1 on all three noise families (82.4%/87.7%/83.8%); held-out-subject AUROC ≈ 0.50 motivates calibration as future work.

TÀI LIỆU

- [1] L. N. Vu, B. T. N. Tran, H. C. Bang, and N. M. Hanh, "Biometric identification through auditory EEG signatures," in *Proc. Int. Conf. Artificial Intelligence for Sustainable Engineering and Innovation (AISEI)*, Irbid, Jordan, 2026, accepted.
- [2] A. van den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, A. Graves, N. Kalchbrenner, A. Senior, and K. Kavukcuoglu, "WaveNet: A generative model for raw audio," arXiv:1609.03499, 2016.
- [3] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," arXiv:2312.00752, 2023.
- [4] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016, pp. 770–778.

- [6] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA, 2019, pp. 4690–4699.
- [7] X. Wang, X. Han, W. Huang, D. Dong, and M. R. Scott, "Multi-similarity loss with general pair weighting for deep metric learning," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA, 2019, pp. 5022–5030.
- [8] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, Boston, MA, USA, 2015, pp. 815–823.
- [9] R. Hadsell, S. Chopra, and Y. LeCun, "Dimensionality reduction by learning an invariant mapping," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, New York, NY, USA, 2006, pp. 1735–1742.
- [10] Y. Luo and N. Mesgarani, "Conv-TasNet: Surpassing ideal time-frequency magnitude masking for speech separation," *IEEE/ACM Trans. Audio, Speech, Lang. Process.*, vol. 27, no. 8, pp. 1256–1266, 2019.
- [11] N. A. Alzahab *et al.*, "Auditory evoked potential electroencephalography-biometric dataset," *Data in Brief*, vol. 57, Art. no. 111065, 2024.
- [12] M. Fallahi, T. Strufe, and P. Arias-Cabarcos, "BrainNet: Improving brainwave-based biometric recognition with Siamese networks," in *Proc. IEEE Int. Conf. Pervasive Computing and Communications (PerCom)*, Atlanta, GA, USA, 2023, pp. 53–60.
- [13] H. Yang, W. Zheng, Y. Zhang, and B.-L. Lu, "EEG authentication based on deep learning of triplet loss," *Neural Network World*, vol. 32, no. 5, pp. 269–283, 2022.
- [14] X. Zhang, L. Yao, S. S. Kanhere, Y. Liu, T. Gu, and K. Chen, "MindID: Person identification from brain waves through attention-based recurrent neural network," arXiv:1711.06149, 2017.
- [15] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, Long Beach, CA, USA, 2017, pp. 5998–6008.
- [16] Y. Gong, Y.-A. Chung, and J. Glass, "AST: Audio spectrogram transformer," in *Proc. Interspeech*, Brno, Czech Republic, 2021, pp. 571–575.
- [17] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "EEGNet: A compact convolutional neural network for EEG-based brain-computer interfaces," *J. Neural Eng.*, vol. 15, no. 5, Art. no. 056013, 2018.
- [18] Y. Wang *et al.*, "EEG-Mamba: A selective state-space model for cross-subject EEG decoding," in *Proc. IEEE Int. Conf. Bioinformatics and Biomedicine (BIBM)*, 2024.
- [19] R. T. Schirrmeister *et al.*, "Deep learning with convolutional neural networks for EEG decoding and visualization," *Hum. Brain Mapp.*, vol. 38, no. 11, pp. 5391–5420, 2017.
- [20] K. Musgrave, S. Belongie, and S.-N. Lim, "PyTorch metric learning," arXiv:2008.09164, 2020.